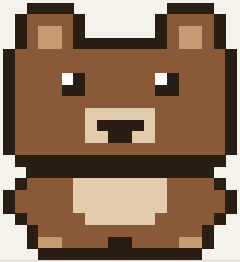
ROMA TRE · A.Y. 2026/27

Policy Evaluation Lab

Lab 1: First steps in R



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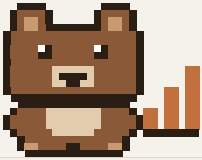
Today

- In Lecture 2 we compared 419 subsidised and 1,081 non-subsidised firms: the naive comparison **overstates** the effect of the grant
- Today we open the same data in R and **reproduce Tables 1 and 2** ourselves
- Plan: RStudio tour → R basics → load and explore the data → naive comparison and balance check → exercises

> **BEFORE WE START**
>
> Download `pel-lab-01.R` and `firms_grant.csv` from the course website (week 2) and put them in the **same folder**

PART 1

R and RStudio



Why R?

- **Free and open source**: you can keep using it after the course, at work or for your thesis
- Used by researchers, central banks (Banca d'Italia, ECB), public agencies and the analytics teams of many firms
- Every step is written in a **script**: anyone can check and rerun your analysis, unlike clicking around in Excel
- Thousands of free packages, including all the methods of this course

> **R** is the engine; **RStudio** is the dashboard we use to drive it. Install both: https://posit.co/download/rstudio-desktop

RStudio: four panes

Top left: script

- where you write your code and save it (`.R` file)
- run a line: **Ctrl+Enter** (Mac: Cmd+Enter)

Bottom left: console

- where R runs the code and prints the results

Top right: environment

- the objects you created (data, results)

Bottom right: files, plots, help

- your folders, your charts, the manual of every function

> **GOLDEN RULE**
>
> Write in the **script**, not only in the console: it is your record of what you did

Working directory

- R reads and writes files in one folder: the **working directory**
- Create a course folder (e.g. `Documents/PEL`) with the script and the data
- In RStudio: *Session* → *Set Working Directory* → *To Source File Location*

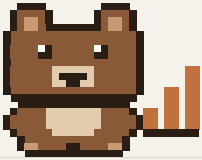
```
getwd() # where am I?
list.files() # which files are here?
```

MOST COMMON ERROR TODAY

`cannot open file 'firms_grant.csv': No such file or directory`
→ the data are not in the working directory

PART 2

R basics



R as a calculator, and objects

```r
2 + 3
sqrt(144)
log(100) # natural logarithm

grant_share <- 0.30 # "<-" gives a name to a value
machinery <- 250000 # cost of new machinery, euro
grant_share * machinery # how much the agency pays: 75000
```

- `<-` creates an **object**: it appears in the Environment pane
- Everything after `#` is a **comment**: R ignores it, you (and your future self) will not
- R is case-sensitive: `Grant` and `grant` are different objects

Vectors and functions

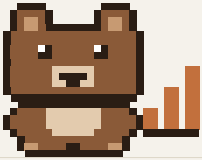
```
growth <- c(11, 3, 8, 6) # c() combines values into a vector
mean(growth) # 7
length(growth) # 4
growth[1] # first element: 11
growth > 5 # TRUE FALSE TRUE TRUE

firm <- c("Alfa", "Beta") # text goes between quotes
?mean # help on any function
```

- A **function** takes inputs (*arguments*) in round brackets and returns a result: `mean(growth)`
- Square brackets select elements: `growth[1]`

PART 3

Our firms



Load and explore the data

```r
firms <- read.csv("firms_grant.csv")

head(firms) # first 6 rows
nrow(firms) # 1500 firms
names(firms) # variable names
summary(firms) # min, mean, max of each variable
View(firms) # spreadsheet view
```

- `firms` is a **data frame**: one row per firm, one column per variable
- One variable: `data$variable`, e.g. `firms$employees_2015`

The variables

Variable	Description
`firm_id`	firm identifier
`region`, `sector`	North / Centre / South; manufacturing sector
`age`	years since foundation (2015)
`family_owned`, `exporter`	1 = yes, 0 = no
`employees_2015`, `sales_2015`	size before the call (sales in thousand euro)
`grant`	**treatment**: 1 = received the grant in 2016
`employees_2019`, `sales_2019`	size after the grant

Simulated data: 1,500 manufacturing SMEs. The outcome of Lecture 2, sales growth, is not in the file: we create it.

Describe one variable at a time

```r
mean(firms$employees_2015) # average size
median(firms$employees_2015) # half of the firms are smaller than...
sd(firms$employees_2015) # standard deviation

table(firms$region) # counts
table(firms$grant) # 1081 without, 419 with the grant
prop.table(table(firms$grant)) # shares
```

QUESTION

Mean and median of employees are quite different. Why? What does it tell you about the size of Italian SMEs?

Create new variables

```r
# Sales growth 2015-2019, in percent: the outcome of Lecture 2
firms$sales_growth <- 100 * (firms$sales_2019 / firms$sales_2015 - 1)
summary(firms$sales_growth)

# A dummy (0/1) for firms in the South
firms$south <- as.numeric(firms$region == "South")
mean(firms$south) # the mean of a dummy is a share
```

- == **compares** (“is equal to?”); <- **assigns**
- The mean of a 0/1 variable is the **share** of firms with value 1

PART 4

Back to Lecture 2

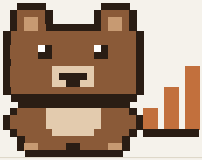


Table 1: the naive comparison

```
mean(firms$sales_growth[firms$grant == 1]) # 9.76
mean(firms$sales_growth[firms$grant == 0]) # 3.37

tapply(firms$sales_growth, firms$grant, mean) # both at once
tapply(firms$sales_growth, firms$grant, sd) # [9.04] and [8.66]
```

- `x[condition]` keeps the elements of `x` for which the condition is TRUE
- `tapply(variable, groups, function)` applies a function within each group

> Subsidised firms grew $9.76 - 3.37 = 6.39$ points faster. Is this difference statistically significant?

The t-test

```
  t.test(sales_growth ~grant, data = firms)
```

```
 Welch Two Sample t-test
t = -12.425, df = 732.8, p-value < 2.2e-16
95 percent confidence interval:
 -7.398839 -5.379820
sample estimates:
mean in group 0 mean in group 1
       3.369519        9.758849
```

- `y ~ group` reads "y by group"
- R computes group 0 minus group 1: the sign is reversed, $|t| = 12.4 > 2.58$
- Significant at 1%. But significant does not mean **causal**: remember Alfa and Beta

Table 2: the balance check

```r
# Were subsidised firms larger BEFORE the grant?
tapply(firms$employees_2015, firms$grant, mean) # 24.65 vs 33.09
t.test(employees_2015 ~grant, data = firms) # |t| = 6.13

# Were they older?
tapply(firms$age, firms$grant, mean) # 24.00 vs 23.63
t.test(age ~grant, data = firms) # |t| = 0.54

tt <- t.test(age ~grant, data = firms) # save the result...
tt$p.value # ...and pick a piece
```

> Same numbers as Lecture 2: subsidised firms are **larger**, but not older

A picture

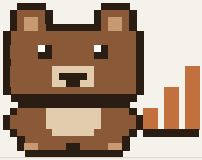
```
hist(firms$sales_growth, breaks = 40,
     main = "Sales growth 2015-2019", xlab = "%")

boxplot(sales_growth ~grant, data = firms,
        names = c("No grant", "Grant"),
        ylab = "Sales growth 2015-2019 (%)")
```

- The plot appears in the bottom-right pane; *Export* saves it as an image for your report
- Look at the overlap: many non-subsidised firms grew faster than many subsidised ones. Averages hide a lot

PART 5

Exercises



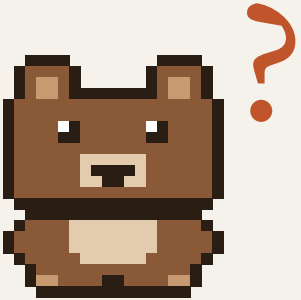
Exercises (in the script, section 8)

1. Create **employment growth** 2015–2019 (%) and compare it between subsidised and non-subsidised firms. Is the difference significant?
2. Complete the **balance check** of Table 2: `sales_2015`, `exporter`, `family_owned`, `south`. Which differences are significant at 1%?
3. Repeat the naive comparison for **exporters only**:
   `exporters <- firms[firms$exporter == 1, ]`
   Does the difference change much? What does it tell you?
4. Which sector has the highest share of subsidised firms?
   `tapply(firms$grant, firms$sector, mean)`

> **NEXT**
>
> Save your script: Lab 2 adds regression (`lm`) to control for these differences

Cheat sheet

Code	What it does
`x <- 5`	create an object
`c(1, 2, 3)`	combine values into a vector
`read.csv("file.csv")`	load a dataset
`head()`, `summary()`, `View()`	look at the data
`data$var`	one variable of a data frame
`mean()`, `median()`, `sd()`, `table()`	describe a variable
`x[cond]`, `data[cond, ]`	keep elements / rows that satisfy a condition
`==`, `!=`, `>`, `<=`, `&`, `|`	compare; and; or
`tapply(y, group, mean)`	a function within each group
`t.test(y ~ group, data = d)`	difference in means and t-test
`?function`	help



Questions?

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